

The Copepod Fauna of New England

Matthew Sasaki

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Preface

Filler text

Acknowledgements

1 How to use this book

This book describes the 53 copepod species currently listed for New England, 3 of which have a full account so far.

Finding a species. Start with the keys (Chapter [4](#)) or the checklist (Chapter [5](#)). Species accounts are grouped by order, then by family. Groups of species that are easily confused have their own pages under *Common comparisons*.

Reading an account. Each account opens with a summary box: authority, classification, habitat, the original description and other key literature, and links to external resources such as WoRMS. It then gives a diagnosis, illustrated morphology, collection records (with a map), and notes. Accounts marked *draft* or *placeholder* are still being written.

Names. Names follow the World Register of Marine Species (WoRMS). Where a name in common use is no longer accepted, the accepted name is given in the summary box.

Terminology. Morphological terms are explained in Chapter [3](#) and the glossary.

2 Collecting and preserving copepods

2.1 Sampling

2.2 Sorting and preservation

2.3 Dissection and imaging

3 Copepod morphology

3.1 Body plan

3.2 Appendages

3.3 The fifth leg (P5)

3.4 Sexes and life stages

4 Identification keys

Warning

Draft. The key to orders below is a starting outline only; check it before use and extend it with keys to genera and species within each order.

4.1 Key to orders of free-living adult copepods

1. Major body articulation behind the fifth pedigerous somite (between it and the genital somite); antennules long, usually 23–25 segments → **Calanoida** ([see part](#))
Major body articulation between the fourth and fifth pedigerous somites → 2
2. Prosome clearly wider than the urosome; antennules of moderate length → **Cyclopoida** ([see part](#))
Body more or less cylindrical, prosome and urosome of similar width; antennules short → **Harpacticoida**

5 Checklist of species

All 53 species currently listed for New England, generated from `data/species.csv`. Species with an account in this book are linked. “Records” is the number of collection records in `data/observations.csv`.

Calanoida

Species	Authority	Family	Habitat
<i>Acartia hudsonica</i>	Pinhey, 1926	Acartiidae	marine
<i>Acartia longiremis</i>	(Lilljeborg, 1853)	Acartiidae	marine
Acartia tonsa	Dana, 1849-1852	Acartiidae	marine
<i>Calanus finmarchicus</i>		Calanidae	marine
<i>Calanus hyperboreus</i>		Calanidae	marine
<i>Calanus minor</i>		Calanidae	marine
<i>Candacia armata</i>	Boeck, 1873	Candaciidae	marine
<i>Centropages bradyi</i>	Wheeler, 1900	Centropagidae	marine
<i>Centropages hamatus</i>	(Lilljeborg, 1853)	Centropagidae	marine
<i>Centropages typicus</i>	Krøyer, 1849	Centropagidae	marine
<i>Pseudocalanus moultoni</i>		Clausocalanidae	marine
<i>Pseudocalanus newmanni</i>		Clausocalanidae	marine
<i>Leptodiaptomus minutus</i>		Diaptomidae	freshwater
<i>Skistodiaptomus pallidus</i>		Diaptomidae	freshwater
<i>Skistodiaptomus pygmaeus</i>		Diaptomidae	freshwater
<i>Eucalanus crassus</i>		Eucalanidae	marine
<i>Eucalanus monachus</i>		Eucalanidae	marine
<i>Eucalanus pileatus</i>		Eucalanidae	marine
<i>Euchaeta norvegica</i>	Boeck, 1873	Euchaetidae	marine
<i>Mecynocera clausi</i>		Mecynoceridae	marine
<i>Metridia longa</i>		Metridinidae	marine
<i>Metridia lucens</i>		Metridinidae	marine
<i>Paracalanus parvus</i>		Paracalanidae	marine
<i>Parvocalanus crassirostris</i>		Paracalanidae	marine
<i>Anomalocera opalus</i>	Pennell, 1976	Pontellidae	marine
<i>Anomalocera pattersoni</i>		Pontellidae	marine

Species	Authority	Family	Habitat
<i>Calanopia americana</i>	Dahl F., 1894	Pontellidae	marine
<i>Labidocera aestiva</i>		Pontellidae	marine
<i>Pontella meadi</i>		Pontellidae	marine
<i>Pseudodiaptomus coronatus</i>		Pseudodiaptomidae	marine
<i>Rhincalanus nasutus</i>		Rhincalanidae	marine
<i>Epischura massachusettsensis</i>		Temoridae	freshwater
<i>Epischura nordenskiöldi</i>		Temoridae	freshwater
<i>Eurytemora affinis</i>		Temoridae	marine
<i>Eurytemora americana</i>		Temoridae	marine
<i>Eurytemora herdmani</i>		Temoridae	marine
<i>Temora longicornis</i>		Temoridae	marine
<i>Temora stylifera</i>		Temoridae	marine
<i>Temora turbinata</i>		Temoridae	marine
<i>Tortanus discaudatus</i>		Tortanidae	marine

Cyclopoida

Species	Authority	Family	Habitat
<i>Corycaeus elongatus</i>		Corycaeidae	marine
<i>Corycaeus ovalis</i>		Corycaeidae	marine
<i>Corycaeus speciosus</i>		Corycaeidae	marine
<i>Corycaeus venustus</i>		Corycaeidae	marine
<i>Diacyclops navus</i>		Cyclopidae	freshwater
<i>Oithona atlantica</i>		Oithonidae	marine
<i>Oithona colcarva</i>		Oithonidae	marine
<i>Oithona plumifera</i>		Oithonidae	marine
<i>Oithona similis</i>		Oithonidae	marine
<i>Oithona spinirostris</i>		Oithonidae	marine
<i>Oncaea venusta</i>		Oncaeidae	marine
<i>Sapphirina auronitens</i>		Sapphirinidae	marine

Harpacticoida

Species	Authority	Family	Habitat
<i>Microsetella norvegica</i>		Ectinosomatidae	marine

Part I

Calanoid copepods

Species of Calanoida recorded from New England. Species with a full account are linked; the list is generated from `data/species.csv`.

Freshwater

- *Leptodiaptomus minutus* (Diaptomidae) — no account yet
- *Skistodiaptomus pallidus* (Diaptomidae) — no account yet
- *Skistodiaptomus pygmaeus* (Diaptomidae) — no account yet
- *Epischura massachusettsensis* (Temoridae) — no account yet
- [Epischura nordenskiöldi](#) (Temoridae)

Marine

- *Acartia hudsonica* (Acartiidae) — no account yet
- *Acartia longiremis* (Acartiidae) — no account yet
- [Acartia tonsa](#) (Acartiidae)
- *Calanus finmarchicus* (Calanidae) — no account yet
- *Calanus hyperboreus* (Calanidae) — no account yet
- *Calanus minor* (Calanidae) — no account yet
- *Candacia armata* (Candaciidae) — no account yet
- *Centropages bradyi* (Centropagidae) — no account yet
- *Centropages hamatus* (Centropagidae) — no account yet
- *Centropages typicus* (Centropagidae) — no account yet
- *Pseudocalanus moultoni* (Clausocalanidae) — no account yet
- *Pseudocalanus newmanni* (Clausocalanidae) — no account yet
- *Eucalanus crassus* (Eucalanidae) — no account yet
- *Eucalanus monachus* (Eucalanidae) — no account yet
- *Eucalanus pileatus* (Eucalanidae) — no account yet
- *Euchaeta norvegica* (Euchaetidae) — no account yet
- *Mecynocera clausi* (Mecynoceridae) — no account yet
- *Metridia longa* (Metridinidae) — no account yet
- *Metridia lucens* (Metridinidae) — no account yet
- *Paracalanus parvus* (Paracalanidae) — no account yet
- *Parvocalanus crassirostris* (Paracalanidae) — no account yet
- *Anomalocera opalus* (Pontellidae) — no account yet
- *Anomalocera pattersoni* (Pontellidae) — no account yet
- *Calanopia americana* (Pontellidae) — no account yet
- *Labidocera aestiva* (Pontellidae) — no account yet
- *Pontella meadi* (Pontellidae) — no account yet
- *Pseudodiaptomus coronatus* (Pseudodiaptomidae) — no account yet
- *Rhincalanus nasutus* (Rhincalanidae) — no account yet

- *Eurytemora affinis* (Temoridae) — no account yet
- *Eurytemora americana* (Temoridae) — no account yet
- *Eurytemora herdmani* (Temoridae) — no account yet
- *Temora longicornis* (Temoridae) — no account yet
- *Temora stylifera* (Temoridae) — no account yet
- *Temora turbinata* (Temoridae) — no account yet
- *Tortanus discaudatus* (Tortanidae) — no account yet

6 *Acartia tonsa*

Authority: Dana, 1849-1852

Classification: Calanoida › Acartiidae

Habitat: marine

Links: [WoRMS](#); [Banyuls copepod database](#)

Collection records in this book: 0

This account is a placeholder; the description is still to be written.

6.1 Diagnosis

6.2 Morphology

6.3 Records

No collection records yet. Add rows to `data/observations.csv`.

6.4 Notes

7 *Epischura nordenskiöldi*

Classification: Calanoida › Temoridae

Habitat: freshwater

Original description: de Guerne and Richard (1889)

Other useful sources: Herrick and Turner (1895); Schacht (1898)

Collection records in this book: 1

This account is a working draft and may change.

7.1 Diagnosis

The female urosome is relatively straight, lacking the torsion seen in other species of *Epischura* (Figure 7.1). The fifth legs of both sexes separate it from *E. massachusettsensis* (Figure 7.2, Figure 7.3; see also Chapter 9).

7.2 Morphology



Figure 7.1: Adult female. Note the relatively straight urosome, lacking the distinctive torsion present in other species of *Epischura*.



Figure 7.2: Adult female P5. The terminal segment bears six spines: one inner, three apical, and two outer. The outer spines are placed in the distal half of the segment, unlike in *E. massachusettsensis*, where spines are distributed along the entire length of the final segment.



Figure 7.3: Adult male P5. The terminal segment of the left leg is blunt and rounded, with four spines distributed along the distal half. The process on the inner surface of the first segment is prominent, with a slight curve. Its length is substantially more than the width of the segment, distinguishing it from the short digitiform process found in *E. massachusettsensis* males.

7.3 Records

Site	State	Coordinates	Date	Temp. (°C)	Stages
Silver Lake (Wilmington)	MA	42.5640, -71.1881	07 Oct 2025	21	adult female; adult male

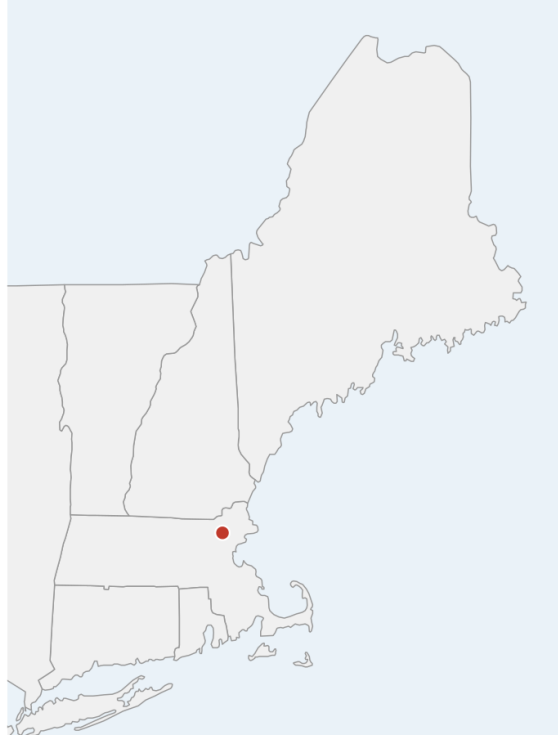


Figure 7.4: Sites where this species has been collected (red) among all sampled sites (grey).

7.4 Notes

Schacht (1898) notes variation in the number of spines on the terminal segment of the female P5. The specimen in Figure [7.2](#) has six spines, but five are also observed.

Part II

Cyclopoid copepods

Species of Cyclopoida recorded from New England. Species with a full account are linked; the list is generated from `data/species.csv`.

Freshwater

- [Diacyclops navus](#) (Cyclopidae)

Marine

- *Corycaeus elongatus* (Corycaeidae) — no account yet
- *Corycaeus ovalis* (Corycaeidae) — no account yet
- *Corycaeus speciosus* (Corycaeidae) — no account yet
- *Corycaeus venustus* (Corycaeidae) — no account yet
- *Oithona atlantica* (Oithonidae) — no account yet
- *Oithona colcarva* (Oithonidae) — no account yet
- *Oithona plumifera* (Oithonidae) — no account yet
- *Oithona similis* (Oithonidae) — no account yet
- *Oithona spinirostris* (Oithonidae) — no account yet
- *Oncaea venusta* (Oncaeidae) — no account yet
- *Sapphirina auronitens* (Sapphirinidae) — no account yet

8 *Diacyclops navus*

Classification: Cyclopoida › Cyclopidae

Habitat: freshwater

Collection records in this book: 0

This account is a placeholder; the description is still to be written.

8.1 Diagnosis

8.2 Morphology

8.3 Records

No collection records yet. Add rows to `data/observations.csv`.

8.4 Notes

Part III

Common comparisons

Side-by-side guides for species (and genera) that are easy to confuse. To add one, copy `comparisons/_template.qmd` to a new file in `comparisons/` (any name not starting with `_`); it is added to the book automatically.

9 *Epischura* species

	<i>Epischura nordenskiöldi</i>	<i>Epischura massachusettsensis</i>
Authority		
Family	Temoridae	Temoridae
Habitat	freshwater	freshwater
Records	1	0

9.1 Key differences

Table 9.1: Characters separating the two New England *Epischura* species.

Character	<i>Epischura nordenskiöldi</i>	<i>Epischura massachusettsensis</i>
Female urosome	Relatively straight, without torsion	<i>To confirm</i> (torsion is present in other <i>Epischura</i>)
Female P5, outer spines of terminal segment	Restricted to the distal half	Distributed along the entire segment
Male P5, process on inner surface of first segment	Long (much longer than segment width), slightly curved	Short and digitiform

References

- de Guerne, Jules, and Jules Richard. 1889. "Révision Des Calanides d'eau Douce." *Mémoires de La Société Zoologique de France* 2: 53–181.
- Herrick, Clarence L., and C. H. Turner. 1895. *Synopsis of the Entomostraca of Minnesota*. Geological and Natural History Survey of Minnesota, Zoological Series 2.
- Schacht, Frank W. 1898. "The North American Centropagidae Belonging to the Genera *Oshranticum*, *Limnocalanus*, and *Epischura*." *Bulletin of the Illinois State Laboratory of Natural History* 5: 225–70.

A Sampling sites

Sites from `data/sites.csv`. Point size shows the number of collection records.

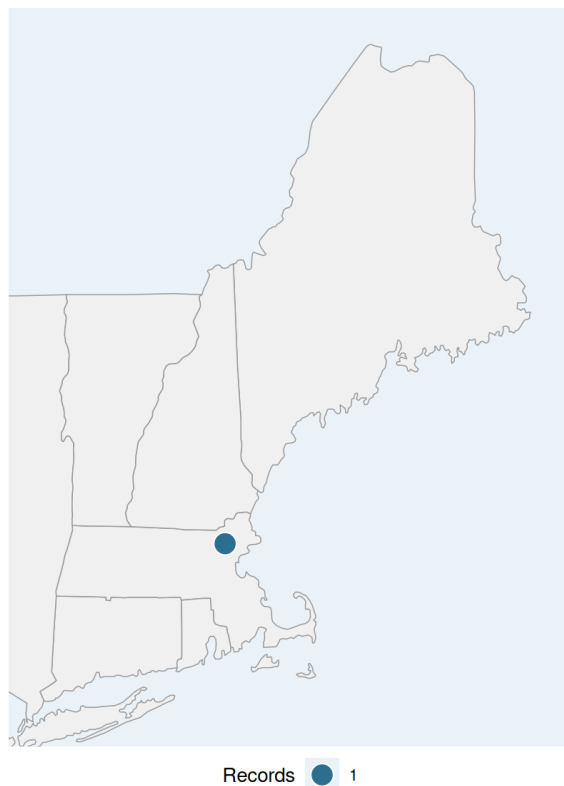


Figure A.1: Sampling sites in New England.

Site	State	Waterbody	Latitude	Longitude	Records
Silver Lake (Wilmington)	MA	lake	42.564	-71.1881	1

B Glossary

Antennule (A1) The first, often long, pair of head appendages.

P5 The fifth leg; often sexually dimorphic and important in identification.

Prosoma / urosoma The anterior and posterior divisions of the body, separated by the major articulation.